What Really Predicts a Hit? Artist History, Era Context and Shortcut Features in Hit Song Prediction

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***Abstract*—Hit song prediction from Spotify audio descriptors has become a popular benchmark, but reported accuracies are rarely compared on the same data, rarely tested for significance, and rarely checked for shortcuts created by the way the datasets were built. We study two simple, non-audio signals: a *leakage-free artist history* feature (the number of Billboard Hot 100 songs the artist placed in strictly earlier periods) and the song's *era*. On the 40,560-song Spotify Hit Predictor dataset, adding both features to the 13 Spotify audio descriptors raises Random-Forest accuracy from 0.782 to 0.832 (+5.0 pts, 25/25 folds of repeated cross-validation, corrected $p = 7 \times 10^{-17}$). In a faithful re-implementation of the only academic study on this dataset, our feature improves its models by +2.1 to +3.2 pts on every one of 50 random splits. Because this benchmark excludes the artists of hits from its negative class, we validate the feature on the peer-reviewed ISMIR 2019 dataset, whose negatives are not selected by artist: it adds +2.4 pts and reaches 0.763, above the best published result (0.750), and it still adds +2.6 pts when hits and non-hits are matched year by year, a control under which release year and decade become useless. These gains hold when no artist appears in both training and test folds, so they do not come from recognising artists seen in training; a cold-start breakdown shows that artist history mainly helps established artists, while era context helps new ones. A systematic audit also shows that several gains are artefacts or noise: song titles (+6.7 pts) encode how the benchmark was assembled, and a 185-dimensional librosa representation that looked significant on a single split (+3.8 pts) is not significant under repeated cross-validation. Spectral descriptors of 30-second previews add +1.4 pts on the whole benchmark, but the gain vanishes on a genre-matched sample: the benchmark's negatives are chosen by genre, and the artist's genre alone lifts accuracy to 0.928. Treated as time series, chart data add nothing before release (market-trend features carry only the era), but once a song has charted its first three weeks predict a top-10 peak with an AUC of 0.89. Lyrics are the only content input that survives every control: on ISMIR they add +4.4 pts with year-matched negatives, English songs only and no artist shared between training and test folds ($p = 3 \times 10^{-5}$). All numbers are reproduced by released scripts.**

***Index Terms*—hit song prediction, music information retrieval, feature engineering, data leakage, shortcut learning, reproducibility**

I. Introduction

Whether a song will become a hit is a question with obvious commercial value and a long, contested research history. Early work concluded that "hit song science is not yet a science" [1], later work argued the opposite on other data [2], and the field has since moved from hand-crafted audio descriptors to convolutional networks on spectrograms [3] and multimodal models combining audio, lyrics and metadata [4]–[6].

A large share of recent studies relies on the audio descriptors exposed by the Spotify Web API and on labels derived from the Billboard Hot 100 chart [7]–[10]. Three weaknesses make these results hard to interpret. First, papers report accuracies on *different* datasets, so numbers such as 86 %, 88 % or 89 % are not comparable. Second, gains are usually measured on a single train/test split without a significance test. Third, the datasets are assembled with sampling rules that a model can exploit: a feature may separate hits from non-hits because of how the negative class was chosen rather than because of anything that makes a song successful, an instance of data leakage [11] and shortcut learning [12].

This paper studies two cheap, non-audio signals—the artist's prior chart history and the song's era—under an evaluation protocol designed to address these three weaknesses. Artist history is not a new idea. Interiano et al. [13] add a "superstar" indicator (whether the artist charted in the recent past) to acoustic features of 500,000 UK releases and find that it raises Random-Forest accuracy, and Middlebrook and Sheik [7] list an artist past-performance field among chart-derived fields, although the inputs of their final model are not stated. Neither study tests the gain for significance, compares it with prior models on the same data, or asks whether it reflects the way the negative class was sampled. What we add is a version restricted to strictly earlier periods, a significance-tested measure of its gain on the same data as prior work, and a check on an independent dataset of whether that gain is a genuine signal or a sampling artefact. Our contributions are:

1) A leakage-free formulation of *artist history* which, together with era, improves every classifier we tested on the Spotify Hit Predictor benchmark (Sec. V-A).

2) A *head-to-head* comparison with the only academic work on that benchmark, using its exact protocol, in which our feature improves its models on 50 out of 50 random splits (Sec. V-C).
3) An *external validation* on the peer-reviewed ISMIR 2019 dataset, including a year-matched control that separates a genuine signal from a sampling artefact, and in which our feature exceeds the best published accuracy (Sec. V-D).
4) A *robustness* analysis showing that the gain is not the model recognising artists it was trained on (artist-grouped cross-validation), together with a cold-start breakdown showing that, for artists with no earlier hit, artist history cannot help and only era does (Sec. V-E).
5) An *ablation* of the final model and a model trained on every input at once, showing where the accuracy comes from (Sec. V-B).
6) An *audit* showing which apparent gains do not survive: title-based features that encode dataset construction, and a rich audio representation whose single-split gain disappears under repeated cross-validation (Sec. V-F), and a genre control showing that a significant spectral-feature gain on the whole benchmark is a genre shortcut (Sec. V-G).
7) *Lyrics*: on ISMIR 2019, the bag of words of a song's lyrics adds +4.4 to +5.1 points and survives year matching, an English-only control and unseen artists, the only content input to do so (Sec. V-H).
8) A *time-series* analysis of the Hot 100: components and forecasting of chart dynamics, market-trend features before release, and early chart trajectories after release (Sec. V-I).

Table I sets each contribution against the closest earlier result.

II. Related Work

Audio-based hit prediction. Pachet and Roy [1] found that acoustic and human-annotated descriptors could not predict popularity, while Ni et al. [2] and Herremans et al. [15] reported above-chance results on chart data. Yang et al. [3] revisited the problem with convolutional networks on audio. Zangerle et al. [4] combined low- and high-level Essentia descriptors [16] of Million Song Dataset tracks [17] in a wide-and-deep network and reached 75.04 % accuracy on a balanced set of 11,664 songs, noting that adding the release year improved every configuration.

Artist reputation. Interiano et al. [13] analyse more than 500,000 UK songs (1985–2015) and show that a superstar indicator improves the prediction of chart success over acoustic features alone. Salganik et al. [18] showed experimentally that success in cultural markets depends on social influence and cumulative advantage, which is one reason why an artist's past success predicts future success.

Spotify descriptors and Billboard labels. Middlebrook and Sheik [7] matched about 24,000 Spotify tracks (1985–2018) to the Hot 100 and reported 87.7 % test accuracy with a Random Forest; their feature list includes chart-derived fields and an artist past-performance feature. SpotHitPy [8] reported about 86 % on roughly 18,500 songs. Zhao et al. [9] added engineered metadata and lyrics features for Top-10 prediction (89.1 % with a Random Forest). These numbers come from different datasets and label definitions. The Spotify Hit Predictor dataset [19] is a public benchmark of decade-balanced hits and non-hits (1960–2019); the only academic study we found on it is an MSc thesis by Du [10], which reports 84 % (2000s) and 82 % (2010s) with one Random Forest per decade.

Multimodal and popularity prediction. Martín-Gutiérrez et al. [5] released SpotGenTrack and predicted Spotify popularity with an end-to-end multimodal network; DeepHits [6] combines log-Mel CNNs, lyric embeddings, artist popularity and release year on the same data; Rompolas et al. [20] use social-media sentiment.

Evaluation pitfalls. Dietterich [21] recommends McNemar's test for a single split, and Nadeau and Bengio [22] show that naive t-tests over overlapping resamples are anti-conservative and propose a corrected variance. Leakage [11] and shortcut learning [12] describe models that exploit cues unrelated to the intended task. In music information retrieval, Sturm [23] showed that systems can reach high scores through confounds in the ground truth ("horses"), and Flexer [24] showed that letting songs by the same artist appear in training and test sets inflates genre-classification accuracy; our artist-grouped cross-validation applies this artist filter to hit prediction. To our knowledge, no hit-prediction study on Spotify descriptors reports both a significance-tested head-to-head on the same data and an audit of construction artefacts.

III. Data

Spotify Hit Predictor. Six decade files (1960s–2010s) list 13 Spotify audio descriptors (danceability, energy, key, loudness, mode, speechiness, acousticness, instrumentalness, liveness, valence, tempo, duration, time signature), two analysis fields (`chorus_hit`, `sections`), the artist, the title and a binary target [19]. After removing duplicate track URIs, 40,560 songs remain. A song is a hit if it appeared on the weekly Hot 100 in that decade. Crucially, the dataset documentation states that a non-hit must satisfy all of: the track is not in that decade's hit list, *its artist is not in that decade's hit list*, it belongs to a "non-mainstream and/or avant-garde" genre, and that genre has no song in the hit list. Each decade file is therefore balanced (48.3–49.9 % hits), and the negative class is selected partly *by artist* and *by genre*. We return to this in Sec. VI.

Audio subset. For 3,562 of these songs a 30-second preview was first retrieved from Deezer (extended to 35,370 songs in Sec. V-F), from which we extracted 31

TABLE I

CONTRIBUTIONS: HOW EACH ONE IS OBTAINED, ON WHICH EXPERIMENT, AND HOW THE RESULT COMPARES WITH THE CLOSEST EARLIER WORK. ACCURACY UNLESS STATED; EARLIER NUMBERS ARE AS PUBLISHED.

Contribution	How we do it	Experiment	Before	Ours
Artist history + era (Sec. V-A)	count of the artist's Hot 100 songs in strictly earlier decades; decade one-hot	Hit Predictor, 40,560 songs, 5×5 CV, 5 classifiers	superstar indicator, no significance test [13]; 13 descriptors: 0.782	0.832 (+5.0, 25/25 folds, $p = 7{\cdot}10^{-17}$); +4.6 to +5.7 for every classifier
Head-to-head (Sec. V-C)	re-implement Du's protocol, add our feature to his models	2000s and 2010s, 50 random 70/30 splits	0.84 (00s), 0.82 (10s) [10]	0.867 and 0.861; better on 50/50 splits
External validation (Sec. V-D)	same recipe on another dataset, with year-matched negatives	ISMIR 2019, 11,662 songs, 5×2 CV	0.750, wide-and-deep, audio + year [4]	0.763; fair test (year-matched) +2.6, $p = 3{\cdot}10^{-5}$
Unseen and new artists (Sec. V-E)	artist-grouped CV; split test songs by earlier hit or not	3 datasets	not tested	gain kept for unseen artists (+5.0, +2.9); new artists: era only (+2.2)
Ablation (Sec. V-B)	remove one input at a time; shuffle one input; train on everything	Hit Predictor, 5×5 CV	not reported	artist −3.0, decade −2.2, instrumentalness −1.6; everything: +5.8, of which −4.0 is the title artefact
Audit of shortcuts (Sec. V-F)	CV instead of one split; genre- and year-matched controls	titles, librosa, FFT, year	gains reported on one split	title +6.7 = artefact; librosa +3.8 → +0.9 n.s.; FFT +1.4 → +0.1
Lyrics (Sec. V-H)	musiXmatch bag of words + 8 lexical features, songs with lyrics only	ISMIR 2019, year-matched, English, unseen artists	lyrics in multimodal models on other data [6], [9]	+4.4 ($p = 3{\cdot}10^{-5}$) under all controls
Early trajectories (Sec. V-I)	first k weekly positions + climb, Random Forest	31,925 Hot 100 songs, top-10 peak	early views predict later views of online content [14]	AUC 0.892 with 3 weeks (0.593 without)

TABLE II

DATASETS. "HIT" DEFINITIONS DIFFER; ACCURACY IS ONLY COMPARABLE WITHIN A ROW.

Dataset	Songs	Hits	Label / negatives
Spotify Hit Predictor [19]	40,560	49.4 %	Hot 100 in decade / rule-based
audio subset (Deezer clips)	3,562	52.0 %	same
ISMIR 2019 [4] (balanced)	11,662	50.0 %	Hot 100 / random MSD
SpotGenTrack [5]	101,909	19.4 %	popularity ≥ 55

spectral/rhythmic descriptors ("FFT") and 185 richer librosa [25] descriptors (20 MFCCs and their first and second derivatives, chroma, tonnetz, spectral contrast, centroid, bandwidth, roll-off, flatness, zero-crossing rate, RMS and onset/tempogram statistics, each summarised by mean and standard deviation).

ISMIR 2019. The release of Zangerle et al. [4] provides Essentia low- and high-level descriptors for 5,832 Hot 100 hits and 89,235 non-hits from the Million Song Dataset. We flatten all numeric descriptors (592 after removing file metadata such as bit rate and codec), keep the 5,831 hits with complete descriptors and, like the original study, draw an equal number of non-hits. Hits have a median release year of 1987 and non-hits of 2001, a fact that matters below.

SpotGenTrack. 101,909 Spotify tracks with 13 descriptors, 207 pre-extracted low-level audio features and a popularity score [5]; we label tracks with popularity ≥ 55 as hits (19.4 %). Because the classes are imbalanced, we report ROC-AUC alongside accuracy for this dataset.

A fourth candidate, a Billboard year-end chart file (2000–2023) augmented with Spotify features, was discarded: 2,911 of its 3,397 rows have no audio descriptors.

IV. METHOD

A. Overview: what goes in, what comes out

Fig. 1 summarises the whole method. The task is binary classification: *input*, one song described by 20 numbers that are all known on the day it is released (13 audio descriptors, six decade indicators and the artist's history); *output*, the probability that the song enters the Billboard Hot 100, turned into a yes/no answer at 0.5. The model is a Random Forest, an ensemble of decision trees that each ask a sequence of threshold questions about the inputs ("is energy above 0.7?", "did the artist have an earlier hit?") and then vote. The same pipeline is run on three datasets, each with its own audio descriptors. Every claimed gain is measured as the difference in held-out accuracy between the pipeline with and without an input, under the protocol in the middle of the figure. A second, separate pipeline (bottom of the figure) answers a different question that can only be asked after release.

B. Features

Audio. The 13 Spotify descriptors (Hit Predictor), the 592 Essentia descriptors (ISMIR) or the 13 Spotify descriptors (SpotGenTrack) form the baseline of each dataset.

Era. On the Hit Predictor we one-hot encode the decade; on ISMIR and SpotGenTrack we use the decade (or the release year) as a number.

Artist history. Let $a(s)$ be the normalised primary artist of song s (accents, punctuation and "feat./&/with"

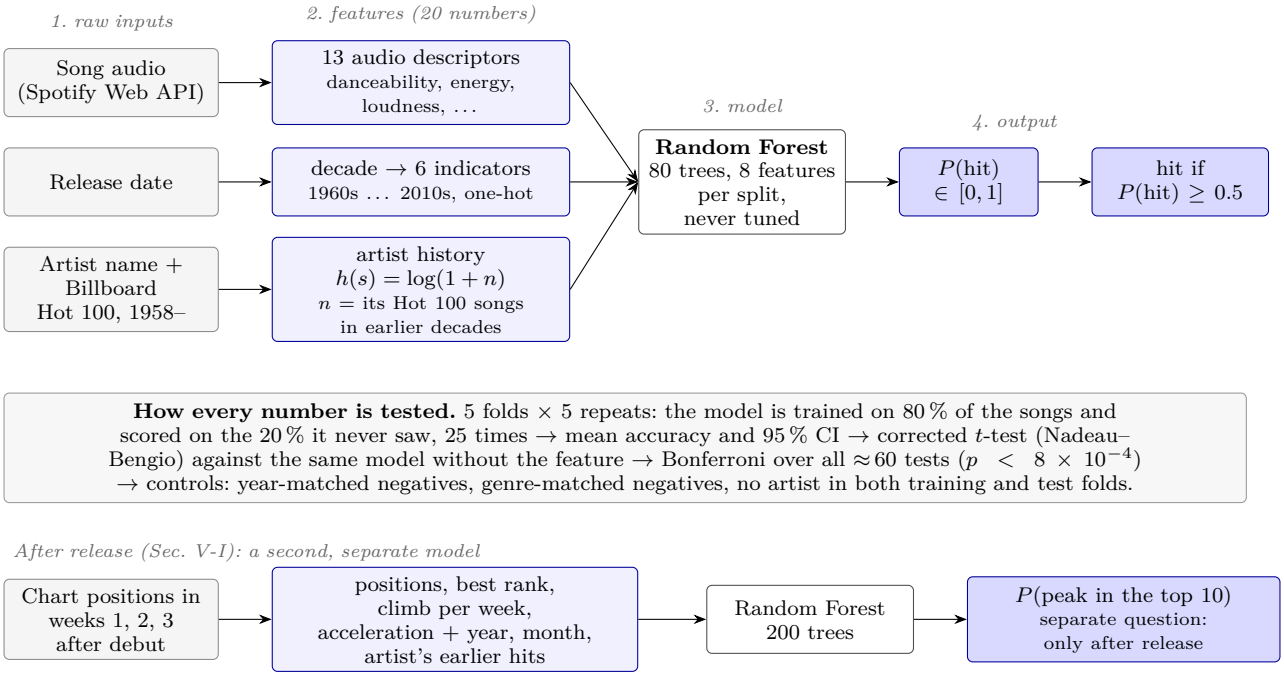


Fig. 1. The prediction pipeline. Top: the final model predicts, on release day, whether a song will enter the Hot 100, from 20 numbers per song. Middle: the protocol under which every gain in this paper is measured. Bottom: a separate model, used only once a song has charted, predicts whether it will reach the top 10 from its first three weeks.

collaborators removed) and $t(s)$ its period (decade on the Hit Predictor, release year elsewhere). With $\mathcal{H}$ the set of reference hits,

$$h(s) = \log\bigl(1 + \bigl|\{\, s' \in \mathcal{H} : a(s') = a(s),\ t(s') < t(s) \,\}\bigr|\bigr). \tag{1}$$

The strict inequality guarantees that no information from the song's own period or from the future is used. In words: for every song, we look up its artist in the full Hot 100 history and count how many different songs that artist had placed on the chart *before* the song's own decade; the logarithm compresses large counts (an artist with 40 earlier hits is not twice as likely to chart again as one with 20). Fig. 2 gives a worked example. The value is computed separately for every artist and every period, so the same artist gets a larger value in each later decade, and an artist with no earlier hit always gets $h(s) = 0$.

On the Hit Predictor, $\mathcal{H}$ is the full weekly Hot 100 history (distinct titles per artist and decade); on ISMIR and SpotGenTrack it is the set of positive songs of the dataset itself. On the Hit Predictor, 52.1 % of songs have an artist that can be matched to the chart history; this includes 99.99 % of the hits, and almost all unmatched songs are flops by artists who never charted (Sec. V-G).

Rich audio. Standardised and reduced by PCA to 40 components; scaling and PCA are fitted on the training part of each fold only.

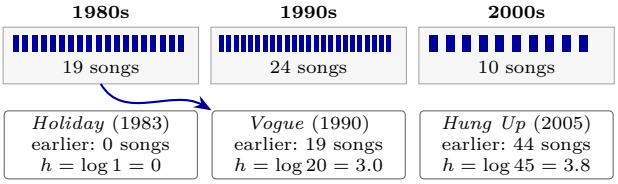


Fig. 2. Artist history, worked example for Madonna. Each tick is one of her Hot 100 songs. For a song from the 1990s (*Vogue*), only her songs from *earlier* decades are counted (the 19 of the 1980s, arrow), never those of the 1990s themselves, so the value is known on release day. A song from her first hit decade gets 0 (*Holiday*), and a 2000s song counts the 1980s and 1990s (44 distinct titles under her normalised name).

C. Models

The main model is the Random Forest [26] configuration of Middlebrook and Sheik [7] (80 trees, 8 candidate features per split, Gini), kept frozen throughout: we do not tune hyperparameters, so that gains come from the features and remain comparable with the published configuration. Robustness checks use XGBoost [27], logistic regression, k-NN and an RBF support vector machine. External datasets use a Random Forest with 200 trees and $\sqrt{d}$ candidate features, as in our original external scripts.

D. Evaluation protocol

We report two estimates for every comparison. The *canonical* estimate reproduces the split of our earlier experiments (seed 42; 20,000 training and 2,000 test songs on the full benchmark, 75/25 elsewhere) and is compared

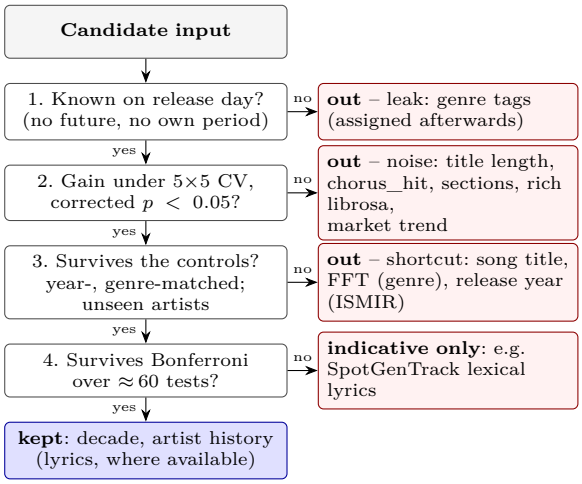


Fig. 3. How every candidate input was kept or rejected. Each input is tested in this order; the boxes on the right list the inputs that stopped at each question.

with McNemar's test [21]. The *cross-validated* estimate uses repeated stratified 5-fold cross-validation (5 repeats unless stated), reports the mean and a 95 % confidence interval, and compares configurations fold by fold with the corrected resampled t-test of Nadeau and Bengio [22]. All learned transformations are inside the cross-validation loop. Title features are produced out-of-fold: a TF-IDF + logistic-regression model is trained within the training part and its predicted probability is used as a single feature.

Two kinds of "new" artist. Two different situations are easy to confuse, and we keep them apart throughout. An *unseen* artist is one whose songs are absent from the training folds; its chart history before the song is still known, so $h(s)$ can be positive. We test unseen artists with artist-grouped cross-validation (Sec. V-E), to check that the model does not simply recognise artists it was trained on. A *new* (cold-start) artist is one with no earlier Hot 100 song, so $h(s) = 0$ by definition, and artist history cannot help them. Our claims that the gain "holds for unseen artists" never mean that it helps new artists; for new artists only era and audio are informative (Table X).

E. How inputs were selected

We did not choose the inputs in advance. Every candidate we built—decade, artist history, song title, title length, `chorus_hit/sections`, a 185-dimensional librosa representation, 31 FFT descriptors, Spotify artist genres, market-trend features and lyrics—went through the same four questions (Fig. 3). We first ask whether the input is known on release day. We then measure its gain under 5×5 cross-validation with the corrected t-test, not on one split. We then re-test it under the controls that remove the known construction shortcuts of each dataset: negatives matched by year (ISMIR), negatives matched by genre (Hit Predictor), and no artist shared between training and test folds. Finally we require significance after a Bonferroni correction over all the tests in the paper. We call an input that passes all four questions a *surviving* input, and we rank all tested gains by this criterion in Table XVII (the *survival ranking*). Only two inputs survive on every dataset: decade (which matters only through its interactions with audio) and artist history. Lyrics survive on the one dataset where we could attach them to both hits and non-hits (Sec. V-H). We also trained a model on everything at once, and we removed one input at a time from both that model and the final model (Sec. V-B).

TABLE III
SPOTIFY HIT PREDICTOR (40,560 SONGS), FROZEN RANDOM FOREST. CANONICAL SPLIT (2,000 TEST SONGS, MCNEMAR) AND 5×5 CROSS-VALIDATION (MEAN, 95 % CI, CORRECTED t-TEST, FOLDS IMPROVED OUT OF 25).

	Canonical		Cross-validation		
Features	Acc.	p	Acc. [95 % CI]	AUC	p (wins)
Audio (13)	.773	–	.782 [.780,.784]	.859	–
+ decade	.796	$3{\cdot}10^{-4}$	.802 [.800,.804]	.880	10^{-10} (25)
+ artist	.800	$6{\cdot}10^{-4}$	.810 [.808,.813]	.900	10^{-11} (25)
+ decade + artist	**.822**	$2{\cdot}10^{-9}$	**.832 [.830,.834]**	**.918**	$7{\cdot}10^{-17}$ (25)
Decade + artist only	.668	–	.670 [.669,.672]	.685	–

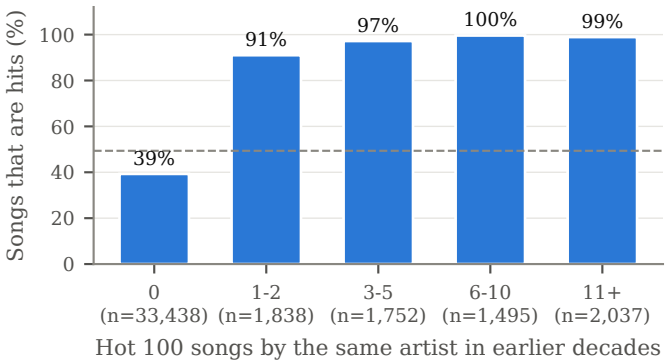


Fig. 4. Share of hits by the number of Hot 100 songs the artist placed in earlier decades (Spotify Hit Predictor, 40,560 songs). Dashed line: overall hit share (49 %).

V. RESULTS

A. Main result on the Spotify Hit Predictor

Table III shows that each feature helps on its own and that their gains are nearly additive. The canonical numbers reproduce our earlier single-split experiments exactly; cross-validation places all accuracies about one point higher and gives very narrow intervals. The combined gain is +5.04 pts and is positive in all 25 folds. It is not bought by trading recall for precision: on the hit class, precision rises from 0.758 to 0.828 and recall from 0.819 to 0.833, so F1 rises from 0.787 to 0.831 (+4.3 pts, 25/25, $p = 7 \times 10^{-15}$). Without audio the two features reach only 0.670, so they complement rather than replace the descriptors. The relation between artist history and the label is strongly nonlinear (Fig. 4): 39.3 % of the 33,438

TABLE IV
CROSS-VALIDATED ACCURACY PER DECADE (RANDOM FOREST).

Features	60s	70s	80s	90s	00s	10s
Audio (13)	.743	.750	.770	.806	.836	.816
+ decade	.764	.763	.794	.840	.851	.833
+ artist	.763	.784	.812	.831	.860	.842
+ decade + artist	**.788**	**.796**	**.836**	**.864**	**.873**	**.868**

TABLE V
ROBUSTNESS TO THE CLASSIFIER (5-FOLD CV, CORRECTED t-TEST; ALL 5 FOLDS IMPROVE).

Classifier	Audio (13)	+ decade + artist	Δ	p
Random Forest [26]	.780	.832	+5.2	$2{\cdot}10^{-5}$
XGBoost [27]	.788	**.845**	+5.7	$3{\cdot}10^{-7}$
Logistic regression	.727	.782	+5.5	$4{\cdot}10^{-5}$
k-NN ($k = 25$)	.750	.796	+4.6	$8{\cdot}10^{-6}$
SVM (RBF)	.773	.828	+5.5	$3{\cdot}10^{-5}$

songs whose artist had no earlier hit are hits, against 91.0 % for one or two earlier hits and 97.2–99.6 % beyond.

The improvement holds in every decade (Table IV), from +3.7 pts in the 2000s to +6.6 pts in the 1980s. Since each decade file is balanced, the decade indicator carries no information on its own; its gain can only come from interactions with the audio descriptors, i.e. from era-specific decision rules.

Robustness to the classifier. The gain is of the same size for five classifiers of very different nature (Table V), so it comes from the features rather than from the model. XGBoost is the strongest on average, but on the canonical split it ties with the frozen Random Forest (0.820 vs 0.822, McNemar $p = 0.87$). TreeSHAP attributions of the XGBoost model rank artist history first (mean $|\phi| = 1.08$), the six decade indicators second (0.86 summed) and instrumentalness third (0.73), ahead of every other descriptor.

Future decades. To test whether the gain survives a temporal shift, we train on all earlier decades and test on a later one, with no song from the test decade seen in training. Artist history improves the Random Forest by +2.5 pts on the 1990s (0.736 → 0.761), +1.9 pts on the 2000s (0.805 → 0.824) and +3.4 pts on the 2010s (0.780 → 0.814; McNemar p between 10^{-7} and 10^{-21}), with similar gains for XGBoost. The decade indicator cannot be used here, since the test decade never appears in training.

B. Where the accuracy comes from

Which descriptors matter: PCA. To see which of the 13 Spotify descriptors carry the signal, we ran a principal component analysis on the standardised descriptors (Fig. 5). The first component (23.6 % of the variance) is an *intensity* axis (energy +0.48, loudness +0.47, acousticness −0.43) and is also the one that best separates hits (ROC-AUC 0.70 on its own). The second (11.1 %) is a *mood* axis (danceability +0.48, valence +0.44, duration −0.36; AUC 0.62). The next four components together explain another 33 % of the variance but separate hits no better than chance (AUC 0.50–0.52), and eight components are needed to reach 81 %. Variance and usefulness therefore do not coincide. Among single descriptors, the most separating are instrumentalness (AUC 0.74, lower in hits), danceability (0.70, higher), loudness (0.64) and valence (0.64); key, duration and tempo are close to chance (0.51–0.52). Replacing the descriptors by their components only loses accuracy: the frozen Random Forest reaches 0.575 with one component, 0.710 with six and 0.765 with all 13, against 0.782 for the raw descriptors (−1.7 pts even though no information is removed, 0/25 folds, $p = 2 \times 10^{-8}$). Trees split on one input at a time, and rotating the axes mixes a sparse, highly informative descriptor (instrumentalness) into components shared with uninformative ones. We therefore keep the raw descriptors; PCA is used only for the 185-dimensional librosa representation (Sec. V-F).

Ablation: remove one input at a time. To see where the power of the final model comes from, we removed each of its inputs in turn and re-ran the 5×5 cross-validation, comparing every reduced model with the full one fold by fold (Fig. 6a). Removing artist history costs the most (−3.0 pts, 0/25 folds better, $p = 7 \times 10^{-13}$), then the decade indicators (−2.2 pts, $p = 5 \times 10^{-13}$), instrumentalness (−1.6 pts, $p = 6 \times 10^{-9}$) and duration (−1.1 pts, $p = 2 \times 10^{-7}$). These four survive the Bonferroni correction. Acousticness, speechiness, energy, danceability and valence each cost 0.3–0.6 pts ($0.004 \leq p \leq 0.04$), because the other descriptors largely carry the same information. Removing key, liveness, time signature, mode or tempo changes nothing ($p \geq 0.30$). Keeping one block alone shows that the blocks are complementary rather than redundant: the 13 descriptors alone reach 0.782, artist history alone 0.670, the decade alone 0.503 (chance, since every decade is balanced), and all three together 0.832. Removing all 13 descriptors at once costs −16.2 pts. So audio carries most of the accuracy, and artist history and era add to it.

Permutation importance. Removing an input lets the model re-learn with the others, so correlated descriptors can stand in for one another. As a second view, we shuffle one input at a time in the held-out folds of the trained final model, which measures how much the model relies on it (Fig. 6b). The ranking is the same at the top: artist history (8.9 pts lost when shuffled), then instrumentalness (6.3), acousticness (4.9), the decade block (4.3), duration (3.9) and danceability (3.4), with key, liveness and time signature near zero. Duration is an instructive case. On its own it barely separates hits (AUC 0.51; Appendix A), but the model relies on it, presumably because song lengths changed across decades and its effect depends on the era. Its value comes from interactions, like that of the decade.

Training on everything at once. One might simply put every input we built into one model. We did this on the 35,370 songs that have a preview (needed for FFT), with

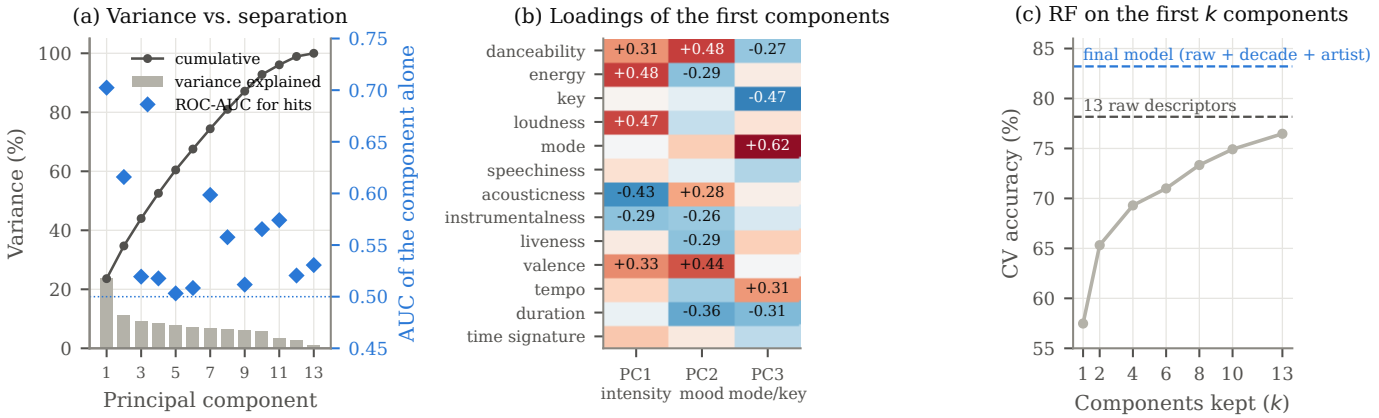


Fig. 5. PCA of the 13 Spotify descriptors (40,560 songs). (a) Variance explained by each component (bars), cumulative (line), and how well each component alone separates hits from non-hits (ROC-AUC, right axis; dotted line = chance). (b) Loadings of the first three components (values below 0.25 in absolute value are not printed). (c) Cross-validated accuracy of the frozen Random Forest trained on the first k components, against the 13 raw descriptors and the final model.

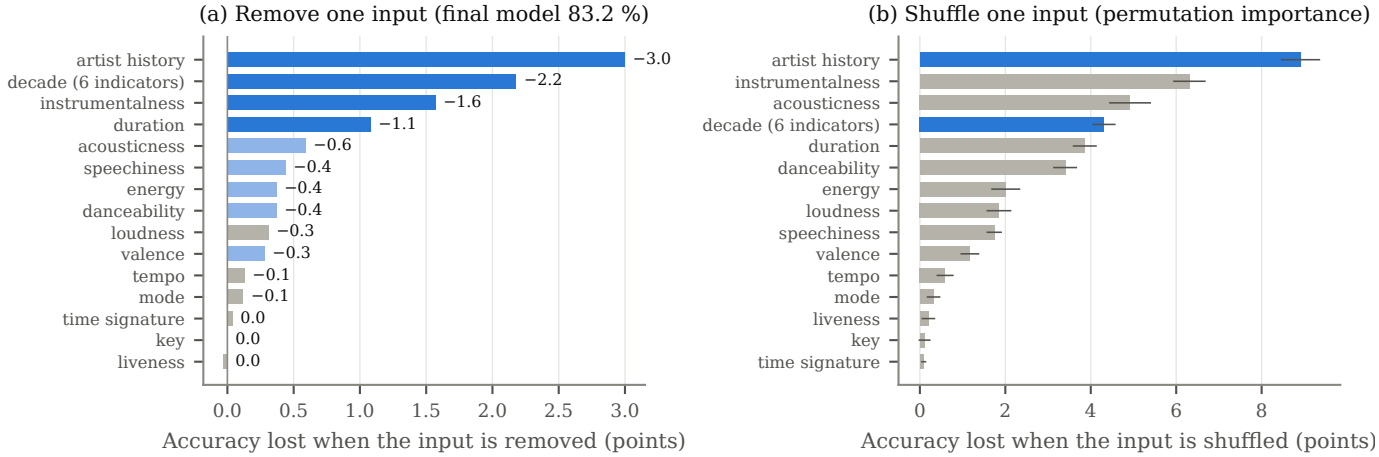


Fig. 6. Where the accuracy of the final model comes from (Hit Predictor, 40,560 songs). (a) Accuracy lost when one input is removed and the model is retrained (5×5 CV; dark blue: significant after Bonferroni, light blue: $p < 0.05$, gray: not significant). (b) Accuracy lost when one input is shuffled in the held-out folds of the trained model (5 folds × 5 shuffles, mean ± s.d.). Blue: non-audio inputs.

the same frozen Random Forest and 5×5 CV (Table VI). The full model reaches 0.883, 5.8 points above the final model (0.825 on these songs). That gain does not come from music. Removing the song title alone takes it down to 0.843 (−4.0 pts, $p = 3 \times 10^{-19}$), and the title is an artefact of how the non-hits were collected (Sec. V-F). The remaining +1.8 pts over the final model comes from FFT and the market-trend features (removing them costs −0.4 and −0.3 pts), whose gains vanish under the genre control and carry only the era (Sec. V-G, V-I). Once FFT and trend are in, the decade indicators become redundant (−0.2 pts, $p = 0.11$): all three describe the era. Artist history remains the input whose removal costs the most among the legitimate ones (−1.9 pts, $p = 2 \times 10^{-11}$), ahead of the 13 descriptors (−1.2 pts), while `chorus_hit`/`sections` and title length change nothing. Training on everything therefore gives a higher number on this benchmark by learning how it was built. The model we keep is the one whose inputs survive the controls: the 20 inputs of the final model.

C. Head-to-head with prior work on the same benchmark

Du [10] trained one Random Forest (450 trees, $m_{\text{try}} = 2$, at most 30 leaves) and one logistic regression per decade on the 2000s and 2010s, with a single 70/30 split. Because the original R random seed cannot be replayed, we repeat the split with 50 seeds. Our re-implementation lands on or slightly above the published figures (Table VII, Fig. 7), which validates the protocol. Adding artist history to *their* models, with *their* inputs, improves accuracy by +2.1 to +3.2 pts on every split, and the improved models exceed the published accuracy on 100 % of the splits. The decade feature is constant within a per-decade model and is therefore not used here.

TABLE VI

TRAINING ON EVERY INPUT AT ONCE (35,370 SONGS WITH A PREVIEW, FROZEN RANDOM FOREST, 5×5 CV). LEFT: ONE BLOCK REMOVED FROM "EVERYTHING" (Δ VS EVERYTHING). RIGHT: THE BLOCK ALONE.

Model	Acc.	AUC	Δ (p)
Final model (Spotify + decade + artist)	.825	.911	–
Everything (8 blocks)	**.883**	.952	+5.8 ($3{\cdot}10^{-21}$) vs final
Everything except song title	.843	.925	+1.8 ($6{\cdot}10^{-10}$) vs final
Remove one block from everything			
– song title	.843	.925	−4.0 ($3{\cdot}10^{-19}$)
– artist history	.864	.933	−1.9 ($2{\cdot}10^{-11}$)
– Spotify (13)	.871	.944	−1.2 ($5{\cdot}10^{-8}$)
– FFT (31)	.879	.949	−0.4 (.01)
– market trend (2)	.879	.949	−0.3 (.01)
– decade (6)	.881	.950	−0.2 (.11)
– title length	.883	.951	−0.0 (.93)
– chorus_hit + sections	.883	.952	+0.0 (.75)
One block alone			
Spotify (13) / FFT (31)	.777 / .773	.849 / .847	–
artist history / song title	.648 / .660	.669 / .740	–
market trend / title length	.629 / .546	.672 / .545	–
chorus_hit + sections / decade	.543 / .532	.563 / .516	–

TABLE VII

HEAD-TO-HEAD WITH DU [10] USING THEIR PROTOCOL (ONE MODEL PER DECADE, 13 AUDIO + `CHORUS_HIT` + `SECTIONS`, 70/30 SPLIT) OVER 50 RANDOM SPLITS: MEAN ACCURACY (STD). "WINS" COUNTS SPLITS WHERE ADDING ARTIST HISTORY HELPS.

Dec.	Model	Published	Re-impl.	+ artist	Wins / p
00s	RF (their cfg.)	.84	.846 (.009)	**.867** (.009)	50 / $9{\cdot}10^{-8}$
	LR (their cfg.)	.82	.821 (.007)	**.848** (.007)	50 / $2{\cdot}10^{-9}$
	RF (frozen)	–	.852 (.008)	**.875** (.006)	50 / 10^{-7}
10s	RF (their cfg.)	.82	.829 (.008)	**.861** (.008)	50 / $4{\cdot}10^{-11}$
	LR (their cfg.)	.79	.807 (.009)	**.835** (.008)	50 / $6{\cdot}10^{-6}$
	RF (frozen)	–	.841 (.008)	**.873** (.008)	50 / $9{\cdot}10^{-10}$

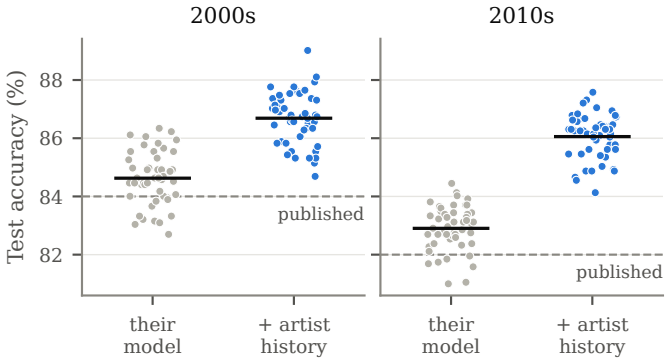


Fig. 7. Head-to-head with Du [10]: accuracy of their Random Forest over 50 random 70/30 splits (dots; bar = mean), without (gray) and with (blue) artist history. Dashed line: their published accuracy.

TABLE VIII

ISMIR 2019 DATA, RANDOM FOREST ON 592 ESSENTIA DESCRIPTORS, 5×2 CROSS-VALIDATION. A: BALANCED RANDOM NEGATIVES (AS IN [4]). B: NEGATIVES MATCHED TO HITS YEAR BY YEAR. p: CORRECTED t-TEST VS AUDIO; ALL 10 FOLDS IMPROVE WHERE SIGNIFICANT.

	A: random negatives		B: year-matched	
Features	Acc. [95 % CI]	p	Acc. [95 % CI]	p
Audio (Essentia)	.708 [.700,.717]	–	.682 [.677,.688]	–
+ year	.739 [.734,.744]	$2{\cdot}10^{-4}$	.681 [.674,.688]	.66
+ artist	.733 [.727,.740]	$2{\cdot}10^{-5}$	**.708** [.702,.714]	$3{\cdot}10^{-5}$
+ year + artist	**.763** [.758,.768]	$7{\cdot}10^{-8}$	**.708** [.702,.715]	$3{\cdot}10^{-5}$
Year alone	.702 [.697,.707]	–	.448 [.444,.451]	–

Best published [4]: 75.04 % (wide-and-deep, audio + year, 5-fold CV)

D. External validation

ISMIR 2019. (Table VIII, Fig. 8.) With random balanced negatives (protocol A, as in the original study), release year alone reaches 70.2 %—as much as 592 audio descriptors—because hits and non-hits come from different eras. Audio + year, which mirrors the inputs of the best model of Zangerle et al., reaches 73.9 %; adding artist history raises it to 76.3 % (+2.4 pts, 10/10 folds, $p = 7{\times}10^{-8}$), above the published 75.04 %. Protocol B removes the era shortcut by drawing, for every hit, a non-hit from the same release year. Year and decade then become useless ($p > 0.5$, and year alone falls below chance), while artist history still adds +2.6 pts ($p = 3 \times 10^{-5}$, 10/10 folds). This is the cleanest estimate of the feature's value: the negatives are not selected by artist and cannot be separated by date.

SpotGenTrack. Here the label is Spotify popularity rather than a chart, and 80.6 % of tracks are negatives, so accuracy is dominated by the majority class: the 13 descriptors score 0.812, barely above always predicting "non-hit". Ranking quality tells the real story (Table IX): artist history raises ROC-AUC from 0.690 to 0.762 (+7.2 points), and with decade to 0.777. On the hit class, the 13 descriptors find only 8.3 % of the hits (F1 0.147); with decade and artist history, recall rises to 25.2 % at unchanged precision (0.62) and F1 to 0.358 (5/5 folds, $p = 7 \times 10^{-7}$), while the 207 low-level audio features add nothing on top of artist history. A single 75/25 split gives the same picture (accuracy 0.810 → 0.822 and ROC-AUC 0.688 → 0.760 with artist history, McNemar $p = 6 \times 10^{-13}$).

E. Unseen artists and cold start

Artist-grouped cross-validation. With random folds, songs by the same artist can fall in both the training and the test part, so a model could recognise an artist from the audio. We therefore repeat every protocol with stratified group k-fold, grouping songs by normalised artist so that no test artist is seen in training (Table X, top). Accuracies drop by up to one point, a small artist-identity effect in the audio descriptors, but the gains remain: +5.0 pts for decade and artist on the Hit Predictor (unchanged, 25/25 folds, $p = 10^{-13}$) and +2.9 pts for artist history in the year-matched ISMIR protocol (10/10, $p = 9 \times 10^{-7}$). In protocol A the gains lose at most half a point, and audio, year and artist still reach 0.759 [0.752, 0.766], above the published 75.04 %.

Cold start. Table X (bottom) splits the grouped test folds by whether the artist had an earlier hit. On the Hit Predictor, artist history does not help new artists

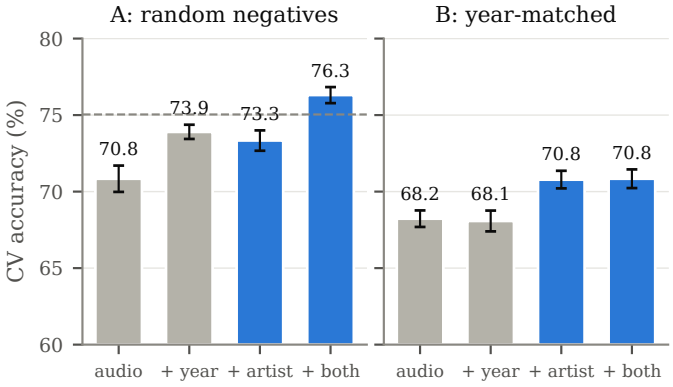


Fig. 8. ISMIR 2019 data, cross-validated accuracy with 95 % CI. Gray: without artist history; blue: with it ("both" = year + artist). Dashed line in A: best published accuracy [4]. Matching years (B) removes the gain of release year but not that of artist history.

TABLE IX
SpotGenTrack (101,909 tracks, 19.4 % hits; always predicting "non-hit" scores 0.806). 5-fold CV, corrected t-test vs the 13 descriptors.

Features	Acc. [95 % CI]	ROC-AUC	p (acc.)
Spotify descriptors (13)	.812 [.811,.813]	.690	–
+ decade	.812 [.811,.813]	.712	.66
+ artist	.823 [.821,.826]	.762	10^{-3}
+ decade + artist	**.825** [.823,.827]	**.777**	$3{\cdot}10^{-4}$
+ artist + low-level audio	.825 [.823,.828]	.758	$2{\cdot}10^{-4}$

(+0.4 pts, $p = 0.17$): its whole gain comes from the 17.6 % of songs by established artists, 96.6 % of which are hits, which quantifies the benchmark effect discussed in Sec. VI. The decade, in contrast, helps new artists (+2.2 pts, $p = 2 \times 10^{-9}$). On ISMIR, where songs by established artists are hits only 72–75 % of the time, artist history also tends to help new artists in the year-matched protocol (+1.4 pts, $p = 0.002$, 10/10 folds), presumably because an empty chart history is itself informative when negatives are not selected by artist.

F. Audit: which gains are not real

Titles. A title model trained out-of-fold adds +6.7 pts, more than any musical feature. Titles do not make songs popular; they reveal the construction of the benchmark, whose non-hits are drawn from niche genres and catalogue releases whose titles differ systematically from Hot 100 singles. We reject this feature despite its significance.

Single split vs cross-validation. On the small audio subset, the 185 rich librosa descriptors appeared to add +3.8 pts with $p = 0.002$ on the canonical split, but the cross-validated gain is +0.9 pts and is not significant. The same holds for the choice of the PCA dimension (5 to 100 components are within noise) and for the knock-out of any descriptor family (all within ±0.1 pts). In a block ablation of the full subset model, only removing artist history causes a significant drop (−3.3 pts, $p = 10^{-5}$).

TABLE X
Robustness. Top: cross-validated accuracy with random stratified folds vs folds grouped by artist (no test artist seen in training). Era = decade (Hit Predictor) or release year (ISMIR). Bottom: gain over audio in grouped CV, on songs whose artist had no earlier hit (new) or at least one (known). New/known songs: 33,438/7,122 (Hit Predictor), 7,480/4,182 (ISMIR A), 7,269/4,393 (ISMIR B).

Data	Folds	Audio	+ era	+ artist	+ both
Hit Predictor	random	.782	.802	.810	.832
	by artist	.772	.792	.802	**.822**
ISMIR A	random	.708	.739	.733	.763
	by artist	.709	.739	.730	**.759**
ISMIR B	random	.682	.681	.708	.708
	by artist	.677	.678	**.705**	**.705**

	Hits (%)	New artists: Δ (p)		Known: Δ (p)
	new/known	+ era	+ artist	+ artist
Hit Predictor	39 / 97	+2.2 ($2{\cdot}10^{-9}$)	+0.4 (.17)	+15.0 ($3{\cdot}10^{-14}$)
ISMIR A	36 / 75	+3.4 ($6{\cdot}10^{-5}$)	+0.8 (.08)	+4.4 (.003)
ISMIR B	37 / 72	+0.1 (.79)	+1.4 (.002)	+5.3 ($8{\cdot}10^{-5}$)

TABLE XI
Audit of candidate gains. Single split (McNemar) vs 5×5 cross-validation (corrected t-test). Top: full benchmark. Bottom: 3,562-song audio subset.

Added to audio	Split Δ (p)	CV Δ	CV p	Verdict
Song title (TF-IDF)	+6.9 ($4{\cdot}10^{-14}$)	+6.7	$4{\cdot}10^{-20}$	artefact
`chorus_hit+sections`	+0.6 (.32)	−0.0	.82	no effect
Title length	+0.05 (1.0)	+0.2	.28	no effect
Rich librosa (PCA 40)	+3.8 (.002)	+0.9	.15	not significant
FFT descriptors (31)	+4.4 (.001)	+2.3	.007	small
Decade	+1.1 (.27)	+1.0	.03	small
Artist	+3.4 (.010)	+3.8	$2{\cdot}10^{-6}$	robust
Artist + rich	+6.7 (10^{-6})	+4.9	10^{-8}	= artist
Decade + artist	+3.6 (.006)	+5.1	10^{-8}	robust

The `chorus_hit` and `sections` fields, often suspected of leakage, add nothing.

FFT and the best audio-subset model. The 31 FFT descriptors are the only extra audio representation whose gain survives cross-validation, and it persists on top of artist history: audio + artist + FFT reaches 0.821 against 0.801 for audio + artist (+2.0 pts, 23/25 folds, $p = 0.008$). Adding decade to that model gives the best subset accuracy, 0.824, but the increment is not significant (+0.3 pts, $p = 0.50$), and decade adds nothing to FFT alone (0.786 vs 0.785): the spectral descriptors already carry the era. Conversely, artist history still adds +3.5 pts on top of FFT ($p = 4 \times 10^{-5}$). Fig. 9 shows every configuration we evaluated on this subset.

FFT on the whole benchmark. We then retrieved Deezer previews for every song we could match: 35,370 of the 40,560 songs (87.2 %). Deezer finds hits more often than flops (94.0 % vs 80.6 %), so the covered set is 53.2 % hits and every comparison below uses the same 35,370 songs. With 5×5 cross-validation, adding FFT to the main model (audio + decade + artist) raises accuracy from 0.825 to 0.839 (+1.4 pts, 25/25 folds, $p = 4 \times 10^{-9}$; AUC

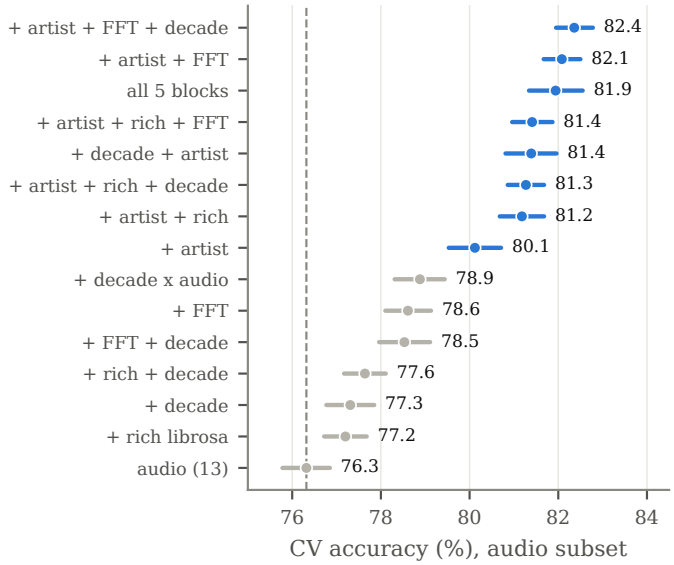


Fig. 9. Every configuration evaluated on the 3,562-song audio subset, 5×5 cross-validated accuracy with 95 % CI, sorted. Blue: with artist history; gray: without. Dashed line: the 13 Spotify descriptors alone.

0.911 → 0.921). The whole gain comes from songs by new artists, 0.789 → 0.807 (+1.8 pts, 25/25, $p = 6 \times 10^{-9}$), while songs by known artists stay at 0.977 (−0.1 pts, n.s.). Unlike on the small subset, decade remains useful once FFT is present: audio + artist + FFT reaches only 0.829 (+0.4 pts over the main model, $p = 0.04$). FFT thus appears to address the cold-start weakness of artist history, but the benchmark's flops come from non-mainstream genres by construction, and spectral descriptors can recognise genre. Sec. V-G shows that this gain does not survive a genre control.

G. Further experiments

Genre control for FFT. We attach Spotify artist genres from the SpotGenTrack release [5] by normalised artist name (20,610 songs with both a preview and a genre) and map the tags to 16 families. The benchmark's construction is plainly visible: Latin, classical, jazz, "world" and metal artists account for 50 % of the flops and 1.9 % of the hits, while rock, pop, R&B, hip-hop and country account for 93 % of the hits. Genre is also partly audible: a Random Forest recovers the family from the 13 Spotify descriptors with 47.4 % accuracy, from the 31 FFT descriptors with 49.9 % and from both with 53.3 % (majority class 26.1 %). Three results follow (Table XII). (a) Genre families as features raise the main model from 0.828 to 0.928 (+10.1 pts, 25/25, $p = 8 \times 10^{-23}$): on this benchmark, knowing the artist's genre is worth more than any audio representation. (b) Once genre is known, FFT *lowers* accuracy (−1.1 pts, 0/25, $p = 5 \times 10^{-7}$), also for new artists (−1.4 pts). (c) On a genre-matched sample (as many hits as flops inside every family; 7,554 songs), FFT adds nothing to the main model (+0.1 pts, 13/25, $p = 0.84$; new artists +0.1 pts, $p = 0.80$). FFT still improves the 13 Spotify descriptors alone on that sample (+2.5 pts, 25/25, $p = 5 \times 10^{-4}$), but the main model, without FFT, still beats audio + FFT by 4.3 pts ($p = 2 \times 10^{-8}$). We conclude that the full-benchmark FFT gain is a genre shortcut, whereas the gain of artist history survives the genre control. Spotify assigns artist genres retrospectively, so we use them only as a control, never as a predictive feature.

TABLE XII
GENRE CONTROL FOR FFT (5×5 CV). LEFT: SONGS WITH A PREVIEW AND A SPOTIFY GENRE. RIGHT: GENRE-MATCHED SAMPLE. ACCURACY OVERALL AND ON SONGS BY NEW ARTISTS.

	All (20,610)		Matched (7,554)	
Model	Acc.	new art.	Acc.	new art.
main (audio+decade+artist)	.828	.780	.790	.749
+ FFT	.841	.798	.791	.750
+ genre	.928	.911	–	–
+ genre + FFT	.917	.897	–	–
FFT vs main, Δ (p)	+1.3 (10^{-4})		+0.1 (.84)	
FFT vs main+genre, Δ (p)	−1.1 (5×10^{-7})		–	

Artist resolution. The 52.1 % match rate is not a matching problem: 99.99 % of the hits are found in the chart history up to their own decade, and the unmatched songs are flops whose artists never charted (2.8 % of flops are matched). Two alternatives give no gain: matching the full credited name before the primary artist ("Simon & Garfunkel" rather than "Simon") lowers accuracy (0.827 vs 0.832, −0.5 pts, $p = 4 \times 10^{-4}$), and crediting featured appearances changes nothing (+0.1 pts, $p = 0.39$). Canonical identifiers would avoid homonyms but cannot raise the match rate.

How fast does a model age? We train on a single decade and test on every later one (Table XIII, songs with a preview). Audio-only accuracy falls from 0.809 within a decade to 0.765 one decade later and 0.689 three decades later. Artist history slows the decay, and its advantage grows with the gap (+2.8 pts within a decade, +5.1 pts three decades later). FFT behaves in the opposite way: it helps within a decade (+2.2 pts) but hurts from three decades on, consistent with descriptors that encode era-specific production and genre rather than lasting appeal. The cells average 6 to 1 train/test pairs and are not tested for significance.

Lyrics on SpotGenTrack. SpotGenTrack provides six lexical descriptors of the lyrics (syllables per word, words per sentence, sentence count, word count, sentence similarity, vocabulary richness) for 79,420 tracks with lyrics (21.5 % hits; 5×1 CV as in Sec. V-D). They improve the 13 Spotify descriptors (AUC 0.678 → 0.729, $p = 3 \times 10^{-4}$) and still add on top of decade and artist history (accuracy 0.808 → 0.814, 5/5 folds, $p = 0.002$; AUC 0.774 → 0.797, $p = 2 \times 10^{-5}$). The gain is modest in accuracy because of the class imbalance, and five folds are too few for it to survive the Bonferroni correction, so we tested lyrics again

TABLE XIII
ROLLING ORIGIN: TRAIN ON ONE DECADE, TEST k DECADES LATER (RANDOM FOREST, 35,370 SONGS WITH A PREVIEW; $k = 0$: 5-FOLD CV INSIDE EACH DECADE). MEAN ACCURACY OVER THE AVAILABLE DECADE PAIRS.

k	pairs	audio	+ artist	+ FFT	+ artist + FFT
0	6	.809	.837	.831	**.859**
1	5	.765	.796	.782	**.810**
2	4	.728	**.775**	.736	**.775**
3	3	.689	**.740**	.673	.728
4	2	.666	**.734**	.661	.718
5	1	.664	**.754**	.599	.706

TABLE XIV
LYRICS ON ISMIR 2019 (SONGS WITH LYRICS ONLY; RANDOM FOREST, 5×2 CV, CORRECTED t-TEST VS THE FINAL MODEL OF THE SAME ROW; ALL ROWS 10/10 FOLDS IMPROVED). UNSEEN: NO ARTIST IN BOTH TRAINING AND TEST FOLDS.

Sample	Songs	Final	+ lyrics	Δ (p)	AUC
A: random	6,526	.728	**.775**	+4.7 ($3{\cdot}10^{-4}$)	.798→.851
B: year-matched	6,478	.676	**.727**	+5.1 ($2{\cdot}10^{-5}$)	.732→.802
A, English	6,364	.718	**.745**	+2.7 ($6{\cdot}10^{-4}$)	.788→.820
B, English	6,294	.658	**.708**	+4.9 (10^{-4})	.720→.772
A, English, unseen	6,364	.712	**.742**	+3.0 ($6{\cdot}10^{-4}$)	.785→.817
B, English, unseen	6,294	.656	**.700**	+4.4 ($3{\cdot}10^{-5}$)	.717→.764

Lyrics alone (B, year-matched): .702, above the final model without lyrics (.676)

on a larger, balanced dataset.

H. Lyrics on ISMIR 2019

ISMIR songs are Million Song Dataset tracks, for which the official musiXmatch release [17] gives the lyrics of 237,662 tracks as word counts over the 5,000 most frequent (stemmed) words. Coverage is itself a shortcut: 56 % of the hits have lyrics against 36 % of the non-hit pool, so "has lyrics" would predict the label. We therefore keep only songs *with* lyrics on both sides and redraw balanced samples under protocols A and B. A second shortcut is language: 97.5 % of the hits but only 80.5 % of the non-hits are mainly in English (at least 25 % of their words among the 29 most common English words). We therefore also repeat every test on English songs only. Two lyrics inputs are added to the final ISMIR model (audio + year + artist history): eight lexical features (log number of words and of distinct words, richness, repetition, share of *I/me/my*, *you/your*, *love* and of common English words), and one bag-of-words score, the out-of-fold probability of a TF-IDF logistic regression trained inside each training fold, as for titles.

Table XIV and Fig. 10a show that lyrics add +4.7 points in protocol A and +5.1 in the year-matched protocol B, in all 10 folds. The gain shrinks but remains when only English songs are kept (+2.7 and +4.9), and it remains when, in addition, no artist appears in both training and test folds (+3.0 and +4.4, $p = 3 \times 10^{-5}$; AUC +0.047, $p = 10^{-6}$), so it does not come from recognising the vocabulary of artists seen in training. These gains survive the Bonferroni correction. In the year-matched protocol, lyrics alone (0.702) are more accurate than audio, year and artist history together (0.676). Fig. 10b shows what the model reads. Hit lyrics favour *love* (by far the largest weight), vocables (*la*, *oh*, *ooh*, *yeah*, *hey*) and direct address (*need*, *girl*, *night*). Non-hit lyrics favour darker or narrative words (*kill*, *hate*, *crime*, *mind*, *their*, *the*). On average, hit lyrics are slightly longer and use a smaller share of distinct words (richness 0.34 vs 0.42), use *love* twice as often (1.3 % vs 0.65 % of words) and use *you* more (5.5 % vs 4.6 %). These words point to themes and genres (romantic pop against metal, punk or hardcore rap) as much as to style. On ISMIR this is a real difference between chart songs and the rest of the catalogue rather than a construction rule, since the negatives are drawn at random. But it means that lyrics, like genre, tell *mainstream* songs from the rest more than they measure the appeal of an individual song. We could not test lyrics on the Hit Predictor, whose songs have no Million Song Dataset identifier.

I. Time series: chart dynamics

The chart as a time series. The weekly Hot 100 (3,545 charts, 1958–2026) also forms time series. The monthly mean number of debuts per chart week (1959–2025) is mostly trend: an STL decomposition [28] gives a trend strength of 0.72 and a seasonal strength of 0.11 [29]. Debuts fall from 11.0 per week in 1959 to 5.5 in 2001 and return to 10.9 in 2025, after the chart started counting streams; the largest residual is a rule change (60 debuts in the chart of 5 December 1998, when songs not released as singles became eligible). The weekly number of holiday songs on the chart is almost purely seasonal: its periodogram peaks at 52.1, 26.1, 17.4 and 13.0 weeks, and a Fourier regression fitted on the streaming era ($K = 20$ harmonics) explains 94 % of its variance. Forecasting monthly debuts 12 months ahead under rolling-origin evaluation (11 origins, 2015–2025), ARIMA$(1, 1, 2)$ [30] and ARIMA with Fourier terms (MASE 1.82) and damped Holt-Winters (1.85) are best and indistinguishable ($p = 0.97$); they beat the seasonal naive forecast (2.32) at 10 of 11 origins ($p = 0.006$). A seasonal ARIMA (1.89) and gradient boosting or random forests on lag and Fourier features (1.94, 1.99) do worse: the series has little seasonality, and 800 points leave tree models little to learn.

Before release: market trends. We describe each Hit Predictor song by the distance of its standardised audio descriptors to the centroid of the previous decade's hits, and by its projection on the direction in which that centroid moved. On top of the main model these two features add +0.07 pts ($p = 0.66$); the 13 per-feature deviations add +0.46 pts (23/25, $p = 0.006$). On audio alone the trend features add +1.8 pts ($p = 7 \times 10^{-10}$), less than the decade alone (+2.0 pts): they carry era information, which the main model already has.

After release: early trajectories. A different question can be asked once a song has entered the chart: will

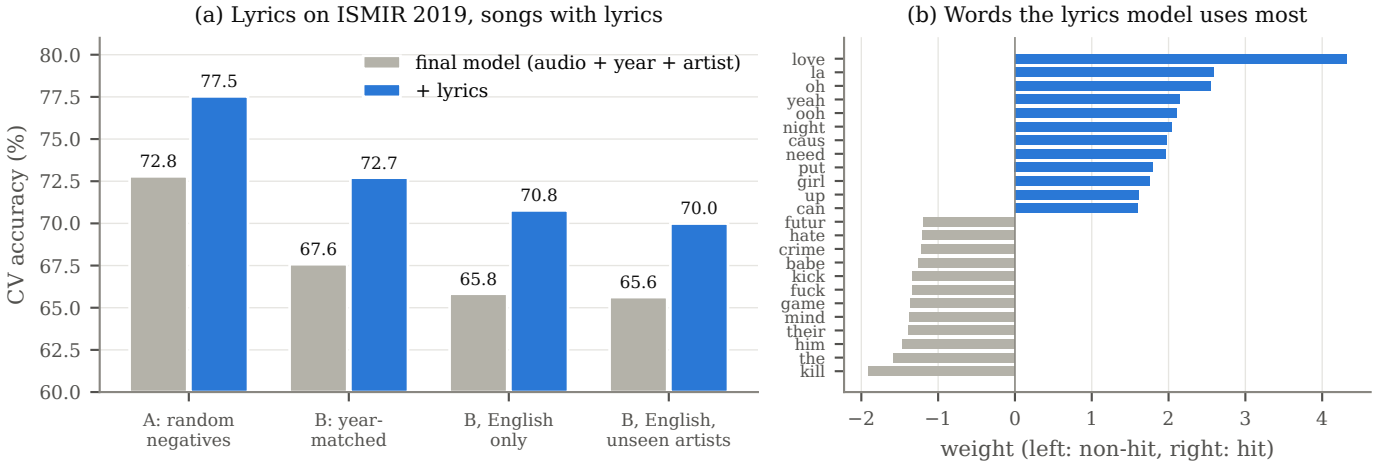


Fig. 10. Lyrics on ISMIR 2019. (a) Cross-validated accuracy of the final model without (gray) and with (blue) the lyrics inputs, from the plain protocol to the strictest control (year-matched, English only, no artist in both training and test folds). (b) Largest weights of the bag-of-words model fitted on the year-matched English sample (stemmed words used in at least 30 songs; "caus" = cause, "futur" = future).

it become a big hit? We take the 31,925 songs with a complete history (debut after the first chart and at least 52 weeks of follow-up; 16.4 % reach the top 10) and describe each by its first k weeks on the chart. Concretely, "the first k weeks" are the k chart positions the song holds in the k weekly charts that start with its debut: week 1 is the debut chart, week 2 the next week's chart, and so on, with position 101 if the song has dropped off. Madonna's *Vogue*, for example, debuted at 39, then held 28 and 12 (so for $k = 3$ the input is $(39, 28, 12)$) and later peaked at number 1. From these positions we also compute the best rank so far, the average climb per week (slope), the change of the weekly move (acceleration) and the number of weeks on the chart, and we add as context the year, month, and the artist's earlier songs and earlier top-10 hits reached before the debut (Random Forest, 200 trees, 5×3 CV). Table XV shows that the trajectory dominates: three weeks give an AUC of 0.892 against 0.593 for the context alone, and context adds +0.03 to +0.05 on top (15/15 folds, $p < 10^{-6}$ for every k). The prediction is not limited to songs already at the top: among the songs not yet in the top 10 after three weeks, the AUC is 0.870, and a model trained on songs before 2000 reaches 0.852 on songs from 2000 onwards. As for online content [14], early popularity predicts later popularity. This task differs from that of Sec. V-A: every song has already charted, so it separates big hits from small ones and cannot be used before release. A second target, at least 20 weeks on the chart, depends on chart rules that remove older songs and is not reported.

What happens in the first three weeks. Fig. 11 shows why so few weeks are enough. Songs that will reach the top 10 already differ at their debut (median position 73 against 87), but the gap opens in the following weeks: their median position climbs to 53 in week 2 and 39 in week 3, while the other songs stay around 80 and one in four has already left the chart by week 3 (Fig. 11a). The climb matters as much as the starting point (Fig. 11b): a song that debuts between 61 and 80 and loses ground reaches the top 10 3 % of the time, but 55 % of the time if it climbs 30 places or more by week 3. Even among debuts in the top 40, the share is 28 % for songs that fall and 82–100 % for songs that climb at least ten places. Permutation importance of the $k = 3$ model (Fig. 11c, held-out AUC 0.893) confirms this. Shuffling the week-3 position costs 0.088 AUC, the climb per week 0.029 and the best rank 0.025, whereas the debut position alone costs 0.004 once the later weeks are known. Among the context inputs, only the year matters (0.023), because chart dynamics changed over time; the artist's earlier songs cost less than 0.01. After release, then, it is the song's own momentum, not the artist's past, that tells a big hit from a small one.

TABLE XV
PREDICTING A TOP-10 PEAK FROM A SONG'S FIRST k WEEKS ON THE HOT 100 (31,925 SONGS, 16.4 % TOP-10; RANDOM FOREST, 5×3 CV). CONTEXT: YEAR, MONTH AND THE ARTIST'S EARLIER CHART HISTORY. LAST COLUMN: SONGS NOT YET IN THE TOP 10 AFTER k WEEKS.

	All songs				Not yet top 10
k	weeks	+ context	F1	context only	+ context
1	.700	.750	.401	.593	.720
2	.796	.850	.531	.593	.828
3	.845	**.892**	.610	.593	.870
4	.891	**.917**	.674	.593	.894

VI. Discussion

The final model. After the audit, the model we retain is the main model of Sec. V-A: the frozen Random Forest (80 trees, 8 candidate features per split) on 20 inputs per song, namely the 13 Spotify descriptors, six decade indicators and artist history $h(s)$. It reaches 0.832 accuracy and 0.918 ROC-AUC on the Hit Predictor (Table III).

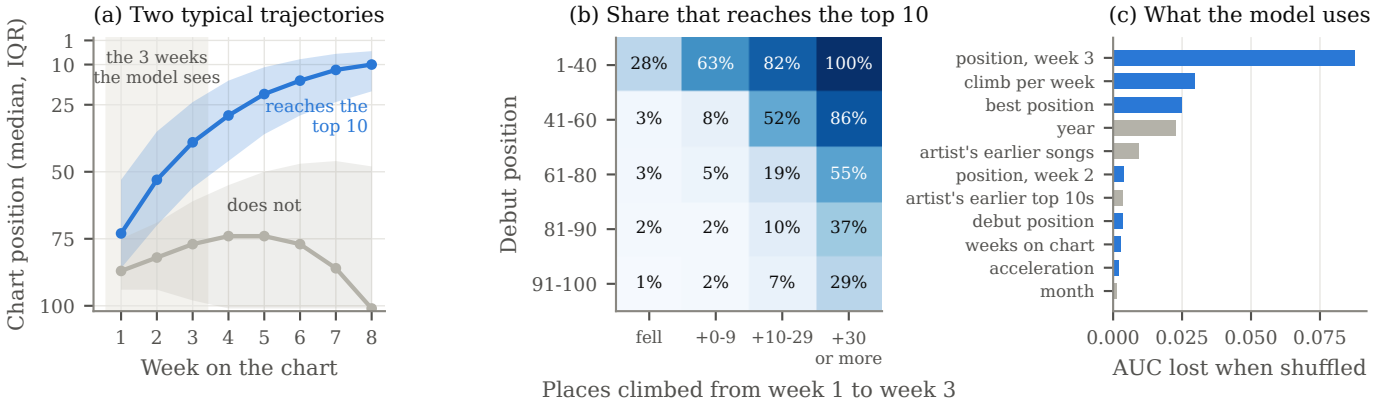


Fig. 11. The first weeks on the Hot 100 (31,925 songs, 5,238 reach the top 10). (a) Median chart position (line) and interquartile range (band) in each week after the debut; 101 = off the chart. The shaded weeks are the inputs of the $k = 3$ model. (b) Share of songs that eventually reach the top 10, by debut position and by places gained between week 1 and week 3 ("fell": lost places or left the chart). (c) Permutation importance of the inputs of the $k = 3$ model on a held-out 25 %: drop in ROC-AUC when one input is shuffled. Blue: trajectory; gray: context.

On the external datasets the same recipe is used with each dataset's own audio descriptors, the release year or decade and artist history, with a 200-tree Random Forest (0.763 on ISMIR protocol A, 0.708 on protocol B, ROC-AUC 0.777 on SpotGenTrack). XGBoost is 1.3 points more accurate under cross-validation but ties with the Random Forest on the canonical split ($p = 0.87$), so we keep the configuration that matches prior work. Every other candidate is left out for a stated reason (Table XVI). A separate model answers a different question, for songs already on the chart: the same kind of Random Forest on the first weeks of the chart trajectory (Sec. V-I). Where lyrics are available (ISMIR, songs with musiXmatch lyrics), adding them gives the best model we found on that dataset (0.775 in protocol A, 0.727 in protocol B; Table XIV).

How much of the gain is real? The benchmark's negative class excludes the artists of that decade's hits. Our feature only counts *earlier* decades, so it is leakage-free in time, but an artist who charted before is likely to chart again and is therefore unlikely to appear as a negative by construction: flops have on average 0.037 earlier hits, against 3.21 for hits. The +5.0 pts on this benchmark, and the head-to-head gains of Table VII, should be read as benchmark-specific. The ISMIR year-matched protocol, whose negatives are chosen neither by artist nor by date, gives a more conservative +2.6 pts; this is the number we consider transferable.

Era as a shortcut. On ISMIR, release year looks like a strong predictor (70.2 % alone, and +12–13 % in the original study) because hits are older than the non-hit pool. Once years are matched, it carries no information. On the Hit Predictor, the decade is uninformative alone and helps only through interactions, which is consistent with era-specific tastes but may also reflect decade-dependent genre rules of the negative class. Reported gains from date features should be accompanied by a year-matched control.

TABLE XVI
INPUTS OF THE FINAL MODEL. GAINS IN ACCURACY POINTS OVER THE 13 SPOTIFY DESCRIPTORS ON THE HIT PREDICTOR (5×5 CV) UNLESS STATED.

Input	Evidence	Decision
13 Spotify descriptors	0.782	kept (baseline)
Decade (6 indicators)	+2.0	kept; helps new artists
Artist history $h(s)$	+2.9; ISMIR B +2.6	kept; survives every control
Song title (TF-IDF)	+6.7	out: construction artefact
Title length	+0.2, n.s.	out: no signal
chorus_hit, sections	0.0, n.s.	out: no signal
Rich librosa (PCA 40)	+0.9, n.s.	out: single-split luck
FFT (31)	+0.1 genre-matched	out: genre shortcut
Spotify artist genre	+10.1	control only: assigned afterwards
Market trend	+0.07, n.s.	out: carries only the era
Lyrics (bag of words + 8 lexical)	ISMIR B +4.4, unseen artists	kept where lyrics exist (ISMIR); unavailable for the Hit Predictor

Genre as a shortcut. On the Hit Predictor, the artist's genre alone takes the main model to 0.928, and the apparent FFT gain disappears once genre is controlled (Sec. V-G). Any audio representation evaluated on this benchmark can score by recognising the excluded genres rather than by modelling what makes a song a hit, so audio gains on it should be reported together with a genre-matched control. Artist history passes that control; FFT does not. Within the audio content itself, therefore, no representation we tested adds to artist history and era once the construction shortcuts are controlled. The one content input that does is the lyrics (Sec. V-H), whose

gain survives year matching, an English-only control and unseen artists; their strongest words, however, suggest that they too partly identify mainstream genres and themes. On these data, what predicts a hit is mostly who sings it, when, and what the song is about—not how it sounds.

Single splits mislead. Several conclusions of our own earlier single-split analysis did not survive cross-validation. With a few thousand songs, an 891-song test set has a standard error of about 1.4 pts, so single-split gains of 3–4 pts on subsets are fragile even when McNemar's test is significant on that split.

Survival ranking. Table XVII lists every gain we tested, split into three groups. *Survived*: low p after Bonferroni and passing the control of its dataset. *Shortcut*: low p but failing the control, so the gain is real on that dataset but would not transfer. *Did not survive*: high p (above the Bonferroni threshold). Only artist history, era, the lyrics and, after release, the early trajectory are in the first group.

Multiple comparisons. The paper reports about 60 p-values without correction for multiplicity. Every result our conclusions rest on remains significant after a Bonferroni correction over all of them ($p < 0.05/60 \approx 8 \times 10^{-4}$): the main gains of Tables III and V, the head-to-head of Table VII, the temporal hold-out, the ISMIR gains of artist history in both protocols, and the artist-grouped gains. The full-benchmark FFT gain on top of the main model also survives ($p = 4 \times 10^{-9}$) but is explained by genre (Sec. V-G); the genre-control results themselves ($p \leq 5 \times 10^{-7}$) survive as well, and so do the early-trajectory gains of Table XV ($p < 10^{-6}$). Results with $8 \times 10^{-4} < p < 0.05$ should be read as indicative only: the decade and FFT gains on the 3,562-song audio subset (including FFT on top of artist history), audio + artist + FFT against the main model on the whole benchmark ($p = 0.04$), the per-feature market-trend deviations ($p = 0.006$), the forecasting comparison of Sec. V-I ($p = 0.006$, 11 origins), the lyrics gains on SpotGenTrack (5×1 CV, $p = 0.002$ for accuracy; the ISMIR lyrics gains, $p \leq 6 \times 10^{-4}$, survive), most single-split gains on that subset, the SpotGenTrack accuracy gain of artist history alone ($p = 10^{-3}$; with decade, $p = 3 \times 10^{-4}$), and the ISMIR cold-start gains of artist history for known artists in protocol A and new artists in protocol B.

Limitations. Artist history is matched by normalised name, not by a canonical identifier, so homonymous artists can be merged. Artist history does not help new artists on the Hit Predictor and helps them only modestly on ISMIR (+1.4 pts), and it partly measures cumulative advantage [18] rather than the intrinsic appeal of a song. We use classical classifiers on fixed descriptors rather than deep networks on raw audio [3], [6]: our question is which inputs carry signal under a controlled protocol, and the comparison with prior work is made at equal model; a learned representation could still capture what the descriptors miss. The comparison with Zangerle et al. uses a classifier rather than their regression on chart position and a different random draw of negatives; the comparison with Du uses re-drawn splits. Our audio previews are 30-second clips from a different service than the one that computed the Spotify descriptors.

VII. Future Work

Sec. V-G already runs the cheap versions of several follow-ups: artist matching (no gain available), a genre control for audio features, rolling-origin decay and lexical lyrics descriptors. What remains needs new data. (i) *Cold start*: artist history barely helps first-time artists on the benchmark, and the FFT gain for them was a genre shortcut; features such as label, collaborators, prior streams or social media remain to be tested, with a genre-matched control. (ii) *Lyrics and emotion*: lyrics for the Hit Predictor songs (matched by title and artist), a genre-matched control for the lyrics gain, semantic and affective lyrics representations, and learned audio embeddings, evaluated with the same audit protocol. (iii) *Benchmark design*: a hit-prediction dataset whose negatives are sampled independently of artist, genre and date. Our genre-matched sample is a small, post-hoc approximation of it; a purpose-built one would let content features be compared without construction shortcuts. (iv) *Early trajectories*: Sec. V-I uses weekly chart positions only; daily streams and radio plays would allow the same prediction days after release.

VIII. Conclusion

A song's artist history is a simple, leakage-free signal that improves hit prediction across classifiers, beats the only prior model on the Spotify Hit Predictor benchmark in a head-to-head comparison, and transfers to an independent peer-reviewed dataset where it exceeds the best published accuracy and survives a year-matched control. These gains hold for artists the model never saw in training, which rules out artist recognition; but artist history cannot help an artist with no earlier hit, and for those newcomers only era context helps; spectral descriptors of a 30-second preview appear to help them too, but that gain disappears under a genre control. The same audit shows that larger gains—song titles, release dates, genre-driven audio features, rich audio on a single split—can come from how the data were built or from chance. Chart time series add nothing before release, but after release a song's first weeks are the strongest predictor we found. The lyrics are the only content input whose gain survives every control. Knowing which gains are real matters more than the headline number.

Code and data availability

Code. https://github.com/REPLACE/hit-song-prediction. Every number in this paper is produced by the scripts in `repro/` (shared data loading and evaluation in `common.py`; one script per section, Table XX) and written to `paper_results/`. Random

TABLE XVII

SURVIVAL RANKING: EVERY TESTED GAIN, RANKED BY HOW FAR IT SURVIVES. A GAIN *SURVIVES* IF IT IS SIGNIFICANT UNDER CROSS-VALIDATION, REMAINS SIGNIFICANT AFTER THE BONFERRONI CORRECTION OVER THE ≈ 60 TESTS OF THE PAPER ($p < 8 \times 10^{-4}$), AND PASSES THE CONTROL THAT REMOVES THE KNOWN SHORTCUT OF ITS DATASET. Δ: ACCURACY POINTS UNLESS STATED; p: CORRECTED t-TEST (MCNEMAR FOR SINGLE SPLITS).

	Gain tested	Δ	p	Bonf.	Control	Verdict
survived (low p)	Decade + artist history, Hit Predictor	+5.0	$7{\cdot}10^{-17}$	✓	✓	real; benchmark-inflated (unseen artists: +5.0, 10^{-13})
	Every classifier, decade + artist (5 models)	+4.6 to +5.7	$\leq 4{\cdot}10^{-5}$	✓	✓	gain comes from the inputs, not the model
	Head-to-head, Du's models + artist (6 settings)	+2.1 to +3.2	$\leq 6{\cdot}10^{-6}$	✓	✓	better on 50/50 splits
	Future decades, artist history (3 decades)	+1.9 to +3.4	$\leq 10^{-7}$	✓	✓	survives a temporal shift
	Decade, Hit Predictor	+2.0	10^{-10}	✓	✓	helps new artists (+2.2, $2{\cdot}10^{-9}$)
	Artist history, ISMIR A (vs audio)	+2.5	$2{\cdot}10^{-5}$	✓	✓	real
	Artist history, ISMIR B year-matched	+2.6	$3{\cdot}10^{-5}$	✓	✓	**transferable estimate**; unseen artists +2.9 ($9{\cdot}10^{-7}$)
	Decade + artist, SpotGenTrack (AUC .690→.777)	+1.3	$3{\cdot}10^{-4}$	✓	✓	F1 .147→.358 ($7{\cdot}10^{-7}$)
	Lyrics, ISMIR B, English, unseen artists	+4.4	$3{\cdot}10^{-5}$	✓	✓	only content input that survives
	Lyrics, ISMIR A, English, unseen artists	+3.0	$6{\cdot}10^{-4}$	✓	✓	same, weaker
	Early trajectory, 3 weeks + context vs weeks (AUC)	+.047	$< 10^{-6}$	✓	✓	after release only; AUC 0.892
	Genre-control results (FFT given genre)	−1.1	$5{\cdot}10^{-7}$	✓	–	shows that FFT encodes genre
shortcut	Song title (TF-IDF), Hit Predictor	+6.7	$4{\cdot}10^{-20}$	✓	×	artefact of how non-hits were collected
	Spotify artist genre, Hit Predictor	+10.1	$8{\cdot}10^{-23}$	✓	×	not known on release day; control only
	FFT, whole Hit Predictor	+1.4	$4{\cdot}10^{-9}$	✓	×	genre-matched: +0.1 ($p = 0.84$)
	Release year, ISMIR A	+3.1	$2{\cdot}10^{-4}$	✓	×	year-matched: −0.1 ($p = 0.66$)
did not survive (high p)	Lyrics lexical, SpotGenTrack (5×1)	+0.6	.002	×	–	indicative; confirmed on ISMIR
	Market trend, 13 per-feature deviations	+0.46	.006	×	–	indicative only
	ARIMA vs seasonal naive (11 origins)	MASE −0.5	.006	×	–	indicative only
	FFT on top of artist, audio subset	+2.0	.008	×	×	genre shortcut (see above)
	Decade, audio subset	+1.0	.03	×	–	small sample
	Audio + artist + FFT vs final, whole benchmark	+0.4	.04	×	×	no
	Rich librosa (PCA 40), audio subset	+0.9	.15	×	–	single split said +3.8 ($p = .002$)
	Artist history for new artists, Hit Predictor	+0.4	.17	×	–	cannot help: $h(s) = 0$
	Title length	+0.2	.28	×	–	no signal
	Market trend, distance + direction	+0.07	.66	×	–	carries only the era
	`chorus_hit` + `sections`	0.0	.82	×	–	no signal

seeds are fixed (42 unless stated). `docs/DATA.md` explains where each dataset goes.

Data. All datasets are public: Spotify Hit Predictor [19], https://github.com/fortyTwo102/The-Spotify-Hit-Predictor-Dataset (also Kaggle: https://www.kaggle.com/datasets/theoverman/the-spotify-hit-predictor-dataset); Billboard Hot 100 weekly charts, https://github.com/utdata/rwd-billboard-data; ISMIR 2019 hit-song data [4], https://zenodo.org/records/3258042; SpotGenTrack [5], https://data.mendeley.com/datasets/4m2x4zngny; musiXmatch lyrics for the Million Song Dataset, http://millionsongdataset.com/musixmatch/. Audio previews were retrieved from the public Deezer API (https://developers.deezer.com/api).

APPENDIX A
INPUTS IN DETAIL

Decade. Six indicators (1960s … 2010s), exactly one equal to 1. On ISMIR and SpotGenTrack, the release year (or decade) as a number.

Artist history. Defined in Sec. IV-B and Fig. 2. Artist names are lower-cased, accents and punctuation are removed, and only the first credited artist is kept (the name is cut at "featuring", "feat.", "ft.", "with", "&", "x", "+", "/", "vs.", "and" or a comma), so that "Madonna feat. Justin Timberlake" counts for Madonna. On the Hit Predictor the reference set is the weekly Hot 100 since August 1958, counted in distinct titles per artist

TABLE XVIII
THE 13 SPOTIFY AUDIO DESCRIPTORS (SPOTIFY WEB API DEFINITIONS) AND HOW WELL EACH ONE ALONE SEPARATES HITS ON THE HIT PREDICTOR (ROC-AUC; SEC. V-B).

Descriptor	Meaning	AUC
instrumentalness	likelihood that the track has no vocals (0–1); lower in hits	.74
danceability	suitability for dancing from tempo, rhythm stability, beat strength (0–1)	.70
loudness	average loudness in dB	.64
valence	musical positiveness, sad (0) to happy (1)	.64
acousticness	confidence that the track is acoustic (0–1); lower in hits	.60
energy	perceived intensity and activity (0–1)	.59
speechiness	presence of spoken words (0–1)	.58
mode	major (1) or minor (0)	.54
time signature	estimated beats per bar	.53
liveness	probability that the track was recorded live	.53
tempo	estimated beats per minute	.52
duration	length in milliseconds	.51
key	pitch class of the track (0 = C, ..., 11 = B)	.51

and decade; on ISMIR and SpotGenTrack it is the set of positive songs of the dataset, counted by release year.

Rejected candidates. *Song title*: TF-IDF of word uni- and bigrams, logistic regression, out-of-fold probability as one feature. *Title length*: number of words. *chorus_hit, sections*: the two Echo Nest analysis fields of the Hit Predictor. *Rich librosa* (185): 20 MFCCs with first and second derivatives, chroma, tonnetz, spectral contrast, centroid, bandwidth, roll-off, flatness, zero-crossing rate, RMS and onset/tempogram statistics, each by mean and standard deviation, reduced to 40 principal components inside each fold. *FFT* (31): from the short-time Fourier transform of the 30-second clip, the mean and standard deviation of the spectral centroid, bandwidth, roll-off, flatness and contrast, 13 mean MFCCs, the chroma mean and standard deviation, and six rhythm descriptors (estimated tempo, onset-envelope mean and standard deviation, pulse clarity, mean and variation of inter-onset intervals). *Genre* (16 families, control only): Spotify artist tags mapped by keyword to hip-hop, R&B/soul, country, rock, metal, electronic, pop, jazz, classical, folk, blues, Latin, reggae, gospel, world, other. *Market trend*: distance of the standardised descriptors to the previous decade's hit centroid, projection on the direction in which that centroid moved, and the 13 per-descriptor deviations.

Lyrics (ISMIR). From musiXmatch word counts: log number of words, log number of distinct words, richness (distinct/total), repetition (count of the most frequent word/total), shares of *I, me, my*, of *you, your*, of *love*, and of the 29 most common English words; plus the out-of-fold TF-IDF logistic-regression probability.

Early trajectory. For k weeks: the k positions (101 = off the chart), the best position, the number of weeks on the chart, the slope $(p_k - p_1)/(k-1)$ (negative = climbing), the acceleration $(p_k - p_{k-1}) - (p_{k-1} - p_{k-2})$; context: debut year and month, log(1 + earlier songs of the artist) and log(1 + earlier top-10 hits that had already reached the top 10 before the debut).

APPENDIX B
MODELS AND SETTINGS

TABLE XIX
EVERY MODEL USED, WITH ALL NON-DEFAULT SETTINGS. NONE WAS TUNED ON THE TEST DATA.

Model	Settings
Random Forest (main)	80 trees, 8 candidate features per split, Gini, fully grown trees, seed 42 (configuration of [7])
Random Forest (external, trajectories)	200 trees, $\sqrt{d}$ candidate features per split
XGBoost	400 trees, depth 6, learning rate 0.05, subsample 0.8, column sample 0.8
Logistic regression	standardised inputs, L2, $C = 1$
k-NN	standardised inputs, $k = 25$
SVM	standardised inputs, RBF kernel, $C = 1$, γ = scale
Du's models [10]	RF 450 trees, $m_{\text{try}} = 2$, at most 30 leaves; logistic regression; one model per decade
Title / lyrics text model	TF-IDF + logistic regression ($C = 4$ titles, $C = 1$ lyrics), inner 5-fold out-of-fold
Rich audio	standardisation + PCA (40) fitted inside each training fold
Cross-validation	repeated stratified k-fold: 5×5 (Hit Predictor), 5×2 (ISMIR), 5×1 (SpotGenTrack), 5×3 (trajectories); stratified group k-fold by artist for the unseen-artist tests
Tests	corrected resampled t-test [22] for CV; McNemar [21] for single splits; Bonferroni over ≈ 60 tests

APPENDIX C
WHICH SCRIPT PRODUCES WHAT

APPENDIX D
TIME-SERIES DETAILS

The chart analysis of Sec. V-I is in a notebook (`timeseries/time_series_hot100.ipynb`) that follows the standard sequence of a time-series study [29]: plot the series, decompose it into trend, seasonality and residual, look for cycles with the periodogram, model the cycles with Fourier terms, test stationarity (ADF, KPSS) and autocorrelation (ACF, PACF, Ljung–Box), and compare forecasting methods by rolling origin.

ARIMA in one paragraph. ARIMA(p, d, q) models a series by differencing it d times until it has no trend, then explaining each value from its own p previous values (autoregression) and from the q previous forecast errors (moving average). We chose $d = 1$ from the stationarity tests and (p, q) by AIC over $p, q \leq 2$. Adding Fourier terms as regressors captures yearly seasonality with few parameters.

TABLE XX
SCRIPTS IN `repro/` AND THE RESULTS THEY PRODUCE (`paper_results/R*_*.csv`).

Script	Results
`r1_kaggle_core.py`	main result, per decade, titles, chorus_hit (Tables III, IV)
`r2_models_time.py`	five classifiers, SHAP, future decades (Table V)
`r3_audio_subset.py`	audio subset, librosa, FFT (Table XI, Fig. 9)
`r4_external.py`	ISMIR, SpotGenTrack (Tables VIII, IX)
`r5_tilburg.py`	head-to-head with Du (Table VII)
`r6_robustness.py`	unseen artists, cold start (Table X)
`r7_fft_full.py`	FFT on the whole benchmark
`r8_future_work.py`	artist matching, genre control, decay, SpotGenTrack lyrics (Tables XII, XIII)
`r9_metrics.py`	precision, recall, F1
`r10_timeseries.py`	market trend, early trajectories (Table XV)
`r11_lyrics_ismir.py`	lyrics on ISMIR (Table XIV, Fig. 10)
`r12_report.py`	PCA, ablation, everything together, trajectory figure (Sec. V-B, Fig. 11)
`make_figures.py`	all data figures
`timeseries/ build_notebook.py`	STL, periodogram, Fourier, ARIMA forecasting (App. D)

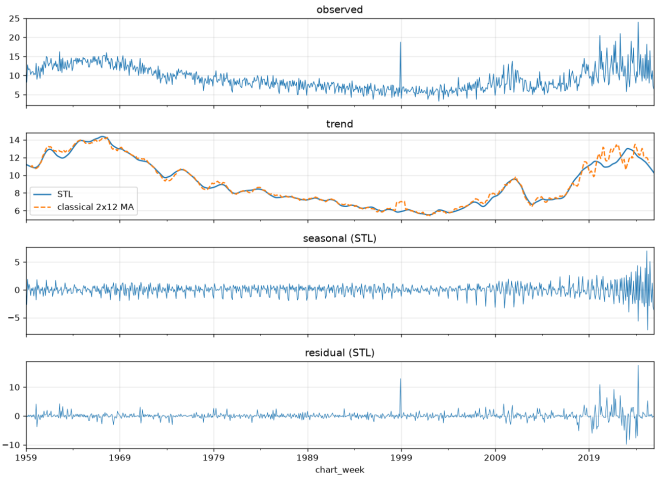


Fig. 12. STL decomposition of the monthly mean number of debuts per chart week (1959–2025). The trend dominates (strength 0.72); seasonality is weak (0.11). The spike of December 1998 is the rule change that admitted songs not released as singles.

APPENDIX E
QUESTIONS A READER MAY ASK

Is artist history not cheating? No information from the song's own period or later is used (Fig. 2), so it is available on release day. What inflates its gain on

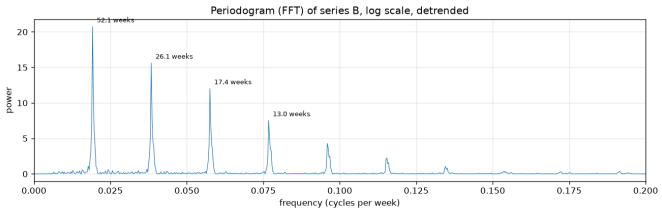


Fig. 13. Periodogram (FFT) of the weekly number of holiday songs on the chart: peaks at one year and its harmonics (26.1, 17.4, 13.0 weeks), which a Fourier regression turns into the seasonal pattern.

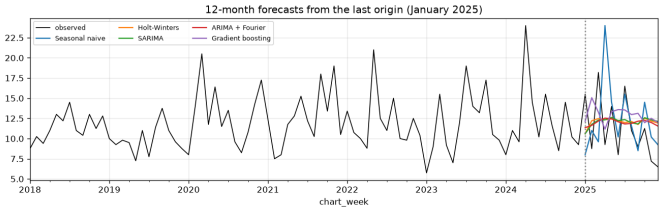


Fig. 14. Twelve-month forecasts of monthly debuts from the last origin (January 2025).

the Hit Predictor is the construction of the negatives (artists of hits are excluded), which is why we report the year-matched ISMIR gain (+2.6 pts) as the transferable number.

Why does the paper say the gain holds for unseen artists but not for new artists? The two words mean different things (Sec. IV-B): an unseen artist is absent from the training folds but may have earlier hits; a new artist has none, so $h(s) = 0$.

Why Random Forest and not XGBoost? XGBoost is 1.3 points better under cross-validation but ties on the canonical split ($p = 0.87$); the frozen Random Forest keeps our numbers comparable with the published configuration, and the gain of our features is the same for all five classifiers (Table V).

Why no deep learning? The question is which inputs carry signal, at equal model. A learned audio representation could capture what the descriptors miss, and is future work.

Why reject the song title when it is the strongest feature? A title does not make a song a hit. It reveals how the non-hits were collected (niche-genre catalogue tracks), so its gain would not transfer to any other dataset.

Why does release year alone score below chance (0.448) in ISMIR protocol B? Every hit has a non-hit from the same year, so the year carries no information, and a Random Forest on a single input with many distinct values fits noise in the training folds; with exactly balanced years, what it learns for a year is then, on average, the opposite of the test fold's labels for that year, which pushes it below 0.5.

Why is FFT rejected although its gain is significant on the whole benchmark? Because it vanishes

TABLE XXI
FORECASTING MONTHLY DEBUTS 12 MONTHS AHEAD, ROLLING ORIGIN (11 ORIGINS, 2015–2025). MASE: MEAN ABSOLUTE ERROR DIVIDED BY THAT OF THE IN-SAMPLE SEASONAL NAIVE FORECAST (LOWER IS BETTER).

Method	MASE
ARIMA$(1, 1, 2)$	**1.82**
ARIMA$(1, 1, 2)$ + Fourier terms	**1.82**
Damped Holt–Winters (ETS)	1.85
Seasonal ARIMA	1.89
Gradient boosting, lags + Fourier	1.94
Random forest, lags + Fourier	1.99
Seasonal naive	2.32

when hits and non-hits come from the same genres (Table XII): spectral descriptors recognise the niche genres from which the benchmark draws its non-hits.

Why are the canonical numbers lower than the cross-validated ones? The canonical split trains on 20,000 songs, cross-validation on 32,448; with more training data every configuration is about one point better, and the differences between configurations are the same.

Why are lyrics kept but not genre, if lyrics also partly reflect genre? Spotify genre tags describe the artist as known today, so they are not known on release day. The words of a song are known on release day. On ISMIR the negatives are random catalogue songs, so the difference in themes between chart songs and the rest is real rather than built into the dataset; we nevertheless state that lyrics measure mainstream-ness more than the appeal of one song (Sec. V-H).

REFERENCES

[1] F. Pachet and P. Roy, "Hit song science is not yet a science," in *Proc. 9th International Conference on Music Information Retrieval (ISMIR)*, 2008, pp. 355–360.

[2] Y. Ni, R. Santos-Rodríguez, M. McVicar, and T. De Bie, "Hit song science once again a science?" in *4th International Workshop on Machine Learning and Music (MML), held with NIPS*, Sierra Nevada, Spain, 2011.

[3] L.-C. Yang, S.-Y. Chou, J.-Y. Liu, Y.-H. Yang, and Y.-A. Chen, "Revisiting the problem of audio-based hit song prediction using convolutional neural networks," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2017, pp. 621–625.

[4] E. Zangerle, R. Huber, M. Vötter, and Y.-H. Yang, "Hit song prediction: Leveraging low- and high-level audio features," in *Proc. 20th International Society for Music Information Retrieval Conference (ISMIR)*, Delft, The Netherlands, 2019, pp. 319–326.

[5] D. Martín-Gutiérrez, G. Hernández Peñaloza, A. Belmonte-Hernández, and F. Álvarez García, "A multimodal end-to-end deep learning architecture for music popularity prediction," *IEEE Access*, vol. 8, pp. 39 361–39 374, 2020.

[6] M. Nofer, V. Nimani, and O. Hinz, "DeepHits: A multimodal CNN approach to hit song prediction," *Machine Learning and Knowledge Extraction*, vol. 8, no. 3, p. 58, 2026.

[7] K. Middlebrook and K. Sheik, "Song hit prediction: Predicting Billboard hits using Spotify data," *arXiv preprint arXiv:1908.08609*, 2019.

[8] I. Dimolitsas, S. Kantarelis, and A. Fouka, "SpotHitPy: A study for ML-based song hit prediction using Spotify," *arXiv preprint arXiv:2301.07978*, 2023.

[9] M. Zhao, M. Harvey, D. Cameron, F. Hopfgartner, and V. J. Gillet, "An analysis of classification approaches for hit song prediction using engineered metadata features with lyrics and audio features," in *Information for a Better World (iConference 2023)*, ser. Lecture Notes in Computer Science. Springer, 2023, pp. 303–311.

[10] J. L. Du, "Spotify between decades: Hit or miss predictor," MSc thesis, Data Science & Society, Tilburg University, 2022, http://arno.uvt.nl/show.cgi?fid=160898.

[11] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, "Leakage in data mining: Formulation, detection, and avoidance," *ACM Transactions on Knowledge Discovery from Data*, vol. 6, no. 4, pp. 1–21, 2012.

[12] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, "Shortcut learning in deep neural networks," *Nature Machine Intelligence*, vol. 2, no. 11, pp. 665–673, 2020.

[13] M. Interiano, K. Kazemi, L. Wang, J. Yang, Z. Yu, and N. L. Komarova, "Musical trends and predictability of success in contemporary songs in and out of the top charts," *Royal Society Open Science*, vol. 5, no. 5, p. 171274, 2018.

[14] G. Szabo and B. A. Huberman, "Predicting the popularity of online content," *Communications of the ACM*, vol. 53, no. 8, pp. 80–88, 2010.

[15] D. Herremans, D. Martens, and K. Sörensen, "Dance hit song prediction," *Journal of New Music Research*, vol. 43, no. 3, pp. 291–302, 2014.

[16] D. Bogdanov, N. Wack, E. Gómez, S. Gulati, P. Herrera, O. Mayor, G. Roma, J. Salamon, J. R. Zapata, and X. Serra, "Essentia: An audio analysis library for music information retrieval," in *Proc. 14th International Society for Music Information Retrieval Conference (ISMIR)*, 2013, pp. 493–498.

[17] T. Bertin-Mahieux, D. P. W. Ellis, B. Whitman, and P. Lamere, "The million song dataset," in *Proc. 12th International Society for Music Information Retrieval Conference (ISMIR)*, 2011, pp. 591–596.

[18] M. J. Salganik, P. S. Dodds, and D. J. Watts, "Experimental study of inequality and unpredictability in an artificial cultural market," *Science*, vol. 311, no. 5762, pp. 854–856, 2006.

[19] fortyTwo102, "The Spotify hit predictor dataset (1960–2019)," GitHub repository, https://github.com/fortyTwo102/The-Spotify-Hit-Predictor-Dataset, accessed Sep. 2026.

[20] G. Rompolas, A. Smpoukis, E. Kafeza, and C. Makris, "Predicting song popularity through machine learning and sentiment analysis on social networks," in *Artificial Intelligence Applications and Innovations. AIAI 2024 IFIP WG 12.5 International Workshops*, ser. IFIP Advances in Information and Communication Technology. Springer, 2024, pp. 314–324.

[21] T. G. Dietterich, "Approximate statistical tests for comparing supervised classification learning algorithms," *Neural Computation*, vol. 10, no. 7, pp. 1895–1923, 1998.

[22] C. Nadeau and Y. Bengio, "Inference for the generalization error," *Machine Learning*, vol. 52, no. 3, pp. 239–281, 2003.

[23] B. L. Sturm, "A simple method to determine if a music information retrieval system is a "horse"," *IEEE Transactions on Multimedia*, vol. 16, no. 6, pp. 1636–1644, 2014.

[24] A. Flexer, "A closer look on artist filters for musical genre classification," in *Proc. 8th International Conference on Music Information Retrieval (ISMIR)*, 2007, pp. 341–344.

[25] B. McFee, C. Raffel, D. Liang, D. P. W. Ellis, M. McVicar, E. Battenberg, and O. Nieto, "librosa: Audio and music signal analysis in Python," in *Proc. 14th Python in Science Conference (SciPy)*, 2015, pp. 18–24.

[26] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[27] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794.

[28] R. B. Cleveland, W. S. Cleveland, J. E. McRae, and I. Terpenning, "STL: A seasonal-trend decomposition procedure based on loess," *Journal of Official Statistics*, vol. 6, no. 1, pp. 3–73, 1990.

[29] R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. Melbourne, Australia: OTexts, 2021.

[30] G. E. P. Box, G. M. Jenkins, G. C. Reinsel, and G. M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed. Wiley, 2015.